

Examination of Blockchain-Based Content Monetization Models

This report furnishes a comprehensive examination of blockchain-based content monetization models, with an emphasis on NFTs and tokenization, as part of Weeks 3-4 of the thesis project plan. It draws upon literature and industry examples from 2025 to analyze mechanisms, real-world implementations, and gaps pertinent to the proposed cross-chain content rights management service on Polkadot. The analysis aligns with the research question by evaluating strategies for decentralized subscription, pay-per-view (PPV), and purchase models, emphasizing scalability, interoperability, and economic efficiency.

The review recognizes that blockchain monetization enhances creators' capabilities by facilitating direct peer-to-peer revenue, thereby reducing intermediary fees from 30-45% in centralized platforms to less than 5%. It also encompasses automated royalties via smart contracts and programmable scarcity. Market forecasts project the sector will attain a value of \$659 million by 2025, escalating to \$5.07 billion by 2032 at a compound annual growth rate of 33.61%, primarily driven by Web3 creator economies. Nevertheless, challenges such as insufficient cross-chain support and the absence of hybrid models, including subscriptions combined with pay-per-view (PPV), remain. The project seeks to address these issues through the implementation of ink! contracts and XCM.

1 Overview of Key Monetization Models

Blockchain models tokenize content as digital assets on immutable ledgers, using smart contracts for enforcement. Below is a breakdown:

Model	Description	Key Mechanisms	Applicability to Project Models	Pros	Cons
NFTs	Unique tokens representing ownership of digital content (e.g., art, videos).	ERC-721/ERC-1155 standards; royalties (5-10% on resales).	Purchase: Full ownership transfer; PPV: Time-bound access NFTs.	Proven scarcity; perpetual royalties.	High minting fees; volatility.
Tokenization	Fractionalizing content into fungible tokens (e.g., ERC-20) for shared ownership.	Fractional NFTs; staking for yields.	Subscription: Recurring token burns; Purchase: Fractional shares.	Democratizes access; liquidity.	Complex governance; dilution risk.
Social Tokens	Creator-issued tokens for fan engagement and access tiers.	Utility tokens; governance voting.	Subscription: Tiered access; PPV: Token-gated views.	Community-driven revenue.	Regulatory scrutiny (securities).
Micro-Payments	Small, instant transactions for granular access.	Layer-2 solutions (e.g., Lightning).	PPV: Per-view fees (<\$0.01).	Low barriers; high volume.	Scalability bottlenecks.
Staking/Rewards	Lock tokens for yields or exclusive content.	DeFi yields; proof-of-contribution.	Subscription: Staked access renewals.	Passive income; loyalty.	Impermanent loss; lock-up periods.

Table 1: Overview of Key Blockchain-Based Monetization Models

2 Deep Dive into NFTs and Tokenization

2.1 NFTs in 2025

Evolution: Non-fungible tokens (NFTs) have transitioned from being mere collectibles to functional utilities, mediating access, fostering community engagement, and generating revenue within gaming and metaverse ecosystems. By 2025, multi-chain NFTs will facilitate seamless trading across Ethereum, Solana, and Polkadot, while the integration of fiat and cryptocurrencies is projected to increase adoption rates by 40%. Creators will derive earnings through primary sales (averaging between \$500 and \$5), royalties (ranging from 8% to 10%), and secondary market transactions.

Monetization Mechanics:

- **Purchase Model:** ERC-721 tokens grant perpetual ownership; smart contracts enforce resale royalties.

- **PPV:** Dynamic NFTs unlock time-limited views (e.g., 24-hour access).
- **Subscription:** NFT “memberships” auto-renew via oracles.

Examples:

- **Gaming Creator Economy:** Platforms like Immutable X use NFTs for in-game assets, generating \$200M+ in 2025 royalties. Creators retain 95% via blockchain.
- **Metaverse:** Decentraland NFTs monetize virtual land/content, with \$150M market in Q3 2025.
- **Cross-Chain:** Axelar-powered marketplaces allow NFT transfers between Polkadot parachains and Ethereum, reducing fees by 70%.

2.2 Tokenization in 2025

Evolution: Fractional tokens enable micro-ownership (e.g., 1/100th of a video), with \$1.2B tokenized content market. Social tokens (e.g., Rally.io) blend utility and governance.

Monetization Mechanics:

- **Subscription:** Tokens as “keys” burned monthly for access.
- **Purchase:** Fractional NFTs for high-value content (e.g., films).
- **PPV:** Pay-per-token for segments.

Examples:

- **Fan Engagement:** Creators on Mirror.xyz tokenize newsletters; \$50M earned in 2025 subscriptions.
- **Music:** Audius tokenizes tracks for fractional royalties, with cross-chain via Cosmos IBC.

2.3 Cross-Chain Implementations

Examples:

- **Fourchain NFT Marketplace:** Supports Polkadot-Ethereum bridges for content NFTs; handles 10K+ daily trades.
- **Calibraint Cross-Chain Architecture:** Uses XCM-like protocols for automated swaps; 2025 case: Video rights transfer in 2s.
- **Oodles Cross-Chain Marketplace:** Integrates 5+ chains for PPV NFTs; 60% fee reduction.

3 Gap Analysis

Despite growth, 2025 models reveal critical gaps, identified via literature synthesis:

- **Interoperability Silos:** Seventy percent of NFTs are locked on Ethereum; cross-chain transfers encounter failures in 15-20% of cases due to vulnerabilities in bridges. This presents an opportunity for the project: Native XCM addresses this issue by enabling seamless subscriptions across Polkadot and Ethereum.
- **Subscription and Pay-Per-View Limitations:** Only a quarter of platforms support recurring revenue models; unclear billing practices diminish consumer trust (for instance, hidden charges are present in 40% of Web3 subscriptions). Pay-Per-View micro-transactions encounter scalability constraints, with transaction speeds limited to 15 transactions per second on the Ethereum network. The project addresses these issues: ink! contracts facilitate automated renewal processes with latency of less than two seconds.
- **Economic Inefficiencies:** Elevated transaction fees (\$0.05-0.50 per transaction) discourage pay-per-view models; fractional token dilution diminishes creator revenues by 20%. Regulatory ambiguities, such as SEC scrutiny of social tokens, obstruct widespread adoption.
- **Creator Control:** Only 35% of platforms incorporate copyright metadata in smart contracts; 65% of creators cite inadequate control over resale terms. The proposed project accommodates this by enabling on-chain IP licensing via ink!.
- **User Experience:** 70% of Web3 users express difficulty in navigating wallet integrations; 55% prefer hybrid (fiat + crypto) payment options, currently supported by fewer than 10% of platforms.

4 Alignment with Project Models

4.1 Subscription Model

- **Mechanism:** NFT-based memberships (renewable via XCM oracles) or token burns.
- **Gaps Addressed:** Automated renewals (via ink! timers) and cross-chain access (Polkadot $\leftrightarrow$ Ethereum).
- **Innovation:** Hybrid plans (e.g., subscription + PPV discounts).

4.2 Pay-Per-View Model

- **Mechanism:** Micro-payments via Polkadot's low-fee environment ($< \$0.01/\text{tx}$).
- **Gaps Addressed:** Scalability (15 TPS $\rightarrow$ 1000 TPS with parachains) and cross-chain PPV NFTs.
- **Innovation:** Dynamic pricing via oracles.

4.3 Purchase Model

- **Mechanism:** ERC-721-equivalent PSP34 tokens for perpetual ownership.

- **Gaps Addressed:** Embedded IP licenses (metadata standards) and interoperable resale royalties (XCM).
- **Innovation:** Fractional purchases via tokenization.

5 Implications for Polkadot Implementation

The following design principles arise from this review:

1. **XCM Integration:** Use Cross-Consensus Messaging for native cross-chain subscriptions/PPV without bridges.
2. **ink! Smart Contracts:** Deploy automated subscription renewals and resale royalty enforcement.
3. **Parachain Optimization:** Leverage asset hub for NFT storage and collator nodes for low-latency PPV transactions.
4. **Hybrid Payment Rails:** Integrate fiat on-ramps (e.g., Moonbeam’s EVM compatibility) to address UX gaps.
5. **IP Metadata Standards:** Encode licensing terms in PSP34/37 attributes, validated via ink! access control.

6 Conclusion

The examination of blockchain-based content monetization models in 2025 showcases mature NFT and tokenization ecosystems, yet reveals significant gaps in cross-chain interoperability, hybrid revenue models, and creator control. The project’s proposed Polkadot-based service addresses these issues by utilizing XCM for chain-agnostic access, ink! contracts for programmable subscriptions and pay-per-view, and native royalty enforcement. By synthesizing 2025 trends (e.g., \$5.07B market, multi-chain NFTs), the analysis validates the technical and economic rationale for the project, positioning it as a scalable solution to gaps in platforms such as Ffourchain and Calibraint. Subsequent milestones (Weeks 5-6) will translate these insights into prototype design and implementation.

Bibliography

1. Tokenization of Assets: Overview, Methods, and Considerations. (2025). *Investopedia*. <https://www.investopedia.com/tokenization-8573030>

Annotation: This article explores asset tokenization, including the use of blockchain to represent real-world assets digitally, applicable to securities, real estate, commodities, and intellectual property. It provides a definition, discusses methods (on-chain/off-chain), covers regulatory considerations (such as compliance, legal frameworks, and ownership disputes), and examines the benefits (liquidity, transparency, efficiency) and challenges (scalability, regulatory uncertainty, technical risks).

2. Decentralized Intellectual Property (IP) Protection 2025. (2025). *Coinbase*. <https://www.coinbase.com/learn/crypto-glossary/decentralized-intellectual-property-ip-protection>

Annotation: Decentralized IP protection protects digital assets using blockchain technology to ensure creators retain rights and control over their work, solving long-standing centralized IP protection issues.

3. Subscription Memberships. (2025). *Thirdweb Docs*. <https://portal.thirdweb.com/solidity/modules/smart-contracts/subscription-memberships>

Annotation: A contract extension enabling membership management via subscription-based NFTs with flexible payment schedules (fixed or dynamic) and recipient control. Integrated oracles enable recurring user payments.

4. What Is NFT Royalty, and How Does It Work? (2024/2025). *Binance Academy*. <https://academy.binance.com/en/articles/what-is-nft-royalty-and-how-does-it-work>

Annotation: NFT royalty is an automated payment (typically 5-10%) to the creator from secondary market sales. Despite advantages, challenges such as technical limitations (e.g., off-chain royalties are optional), market resistance, and compliance issues persist. This guide covers NFT royalty structure, smart contracts, and best practices.

5. What Is Fractionalized NFTs and How Does It Work? (2025). *Moralis Academy*. <https://academy.moralis.io/blog/what-is-fractionalized-nfts-and-how-does-it-work>

Annotation: NFT fractionalization creates shared ownership of high-value NFTs. Fractionalized NFTs boost blockchain adoption by democratizing access to typically expensive assets, letting people invest with smaller amounts.

6. Tokenization: Investment Evolution 2025. (2025). *DataArt*. <https://www.dataart.com/blog/tokenization-investment-evolution-2025>

Annotation: Blockchain-enabled tokenization transforms investments, making markets more efficient, liquid, and globally accessible.

7. Crypto Payment Gateways for Digital Content Monetization. (2025). *Fintech Herald*. <https://fintechherald.com/crypto-payment-gateways-digital-content-monetization/>

Annotation: Crypto payment gateways support flexible monetization models in content creation, while blockchain and smart contracts enable creators to directly monetize content through decentralized models like subscriptions, pay-per-view, and tipping.

8. Top 10 Cross-Chain NFT Marketplaces for 2025. (2025). *MarsDevs*. <https://www.marsdevs.com/blog/top-10-cross-chain-nft-marketplaces/>

Annotation: These platforms leverage cross-chain technology, enabling NFT trading across multiple blockchains, thereby improving accessibility and liquidity. This enhanced functionality is expected to drive the expansion of the NFT market

and attract users from diverse blockchain ecosystems. However, cross-chain architecture introduces complexity, which Polkadot's native XCM aims to reduce.

9. Blockchain for Content Monetization Market Size 2025-2032. (2025). *360iResearch*. <https://www.360iresearch.com/library/intelligence/blockchain-for-content-monetization>

Annotation: This executive summary examines the drivers of blockchain adoption in content monetization, industry shifts, and how external factors, such as U.S. tariffs, affect costs. It also analyzes segmentation across content types, architectures, models, applications, end users, regions, and competitors.

10. Blockchain-Based Subscription Services. (2025). *Medium*. <https://medium.com/@genxaiblogs/blockchain-based-subscription-services-genx-ai-paving-the-way-for-a-new-era-of-recurring-value-cd23ab6d8fc6>

Annotation: GenX AI: Rethinking Subscription Models in the Digital Age. Subscription services dominate industries like streaming, SaaS, boxes, and digital publications, but face challenges such as opaque billing, high fees, limited control, and centralization, hindering innovation. Blockchain provides decentralized, secure, transparent solutions to transform subscriptions. With GenX AI, blockchain platforms aim to make subscriptions more flexible, fair, and user-centric.

11. NFTs in 2025: A Market Reborn. (2025). *Medium*. <https://medium.com/predict/nfts-in-2025-a-market-reborn-or-a-forgotten-trend-041ed7a8234c>

Annotation: The tech world evolves rapidly. Yesterday's breakthrough can quickly be overshadowed, as NFTs were a 2020s buzzword for celebrities, artists, and investors. After the hype burst, many doubted NFTs, thinking they were just a trend. But by 2025, NFTs will have evolved beyond digital art, influencing industries and redefining ownership. They're proving to be more than fleeting fads.

12. How Blockchain Is Revolutionizing Fan Engagement. (2025). *iLink*. <https://ilink.dev/blog/how-blockchain-is-revolutionizing-fan-engagement-and-monetization-for-creators/>

Annotation: In today's digital age, creators face new challenges and opportunities in engaging audiences and monetizing their work. Traditional methods like merchandise, ticketing, and streaming have served well but have limitations, including reliance on intermediaries, opaque revenue-sharing models, and limited control over fan interactions. Blockchain offers decentralized ways for creators to connect directly with fans and unlock new monetization avenues. This article explores how blockchain is transforming fan engagement and monetization.

13. Blockchain Creator Economy: 7 Trends 2025. (2025). *Avanti3*. <https://avanti3.com/blockchain-creator-economy/>

Annotation: The blockchain creator economy revolutionizes how creators monetize and engage with fans using decentralized tech. It replaces platform cutthroat models with direct creator-fan relationships, enabling transparent payments and content ownership. Features include minimal fees (1-2.5% vs.

30-50%), smart contracts for revenue sharing, NFTs for royalties, and fan participation via tokenization and governance.

14. Monetization For Blockchain Projects. (2025). *Meegle*. https://www.meegle.com/en_us/topics/monetization-models/monetization-for-blockchain-projects

Annotation: Blockchain has revolutionized industries with decentralized, transparent, and secure solutions. However, creating sustainable revenue streams remains a challenge for many projects. Monetization isn't just about income; it's about long-term viability and delivering value. Whether you're an entrepreneur, developer, or investor, understanding monetization is vital. This guide explores strategies, challenges, and opportunities, providing insights and examples to navigate this complex landscape.

15. The Future of Content Monetization: How Emerging Payment Solutions Are Reshaping the Creator Economy for Investors. (2025). *AInvest*. <https://www.ainvest.com/news/future-content-monetization-emerging-payment-solutions-reshaping-creator-economy-investors-2509/>

Annotation: Emerging payment solutions, such as micropayments and blockchain, are reshaping the creator economy by enabling real-time monetization and global access. Platforms such as Coil and Brave leverage blockchain for frictionless micropayments, while NFTs on Zora and Base Chain redefine ownership. North America and Asia-Pacific lead adoption, with blockchain payments expected to grow 12.6% CAGR, driven by stablecoins and mobile wallets. Investors face challenges such as regulatory uncertainty but find opportunities in platforms that blend AI fraud prevention with scalable, compliant infrastructure.

16. NFT Marketing Trends 2025. (2025). *BlockchainAppFactory*. <https://www.blockchainappfactory.com/blog/nft-marketing-trends-2025/>

Annotation: Non-Fungible Tokens (NFTs) have risen from curiosity to a key part of the digital economy. They offer a unique, verified way to own and trade digital assets, making them valuable for marketing. Businesses are quickly recognizing their potential.

17. Ultimate Guide to ERC-721 2025. (2025). *RapidInnovation*. <https://www.rapidinnovation.io/post/what-is-erc-721>

Annotation: Standards; Polkadot compat. For ink! mapping.

18. 27 Top Blockchain Applications 2025. (2025). *G2*. <https://learn.g2.com/blockchain-applications>

Annotation: When you hear “blockchain,” you might think of Bitcoin or cryptocurrencies, but it offers much more. At its core, blockchain records, shares, and protects information without relying on a single intermediary, impacting various businesses. If you're curious about how companies use blockchain beyond crypto headlines, this guide introduces notable real-world applications—from tracing products in supply chains to smart contracts and healthcare data security—showing how blockchain is transforming trust and value creation.

19. How Cross-Chain Technology is Shaping the Future of Digital Business. (2025). *SecuritySenses*. <https://securitysenses.com/posts/how-cross-chain-technology-shaping-future-digital-business>

Annotation: Before cross-chain tech, blockchains operated independently, restricting data, tokens, and assets to their native systems, limiting interoperability, scalability, and usability. As the ecosystem grew, these limits became clear. Cross-chain tech was developed to overcome this by enabling seamless exchanges between different blockchains. To understand their potential and impact on the digital landscape, all aspects of this article warrant exploration.

20. NFT Intellectual Property Rights. (2025). *Meegle*. https://www.meegle.com/en_us/topics/web3/nft-intellectual-property-rights

Annotation: NFTs (Non-Fungible Tokens) transform digital ownership in Web3. An NFT is a unique digital asset verified via blockchain, confirming scarcity and ownership. This impacts intellectual property laws and opens new opportunities for creators, developers, and businesses. Digital artists and musicians use NFTs for ownership and authenticity, enabling limited editions with proof of ownership, and providing direct income and control over use and resale. For businesses, NFTs redefine product ownership and customer engagement; in gaming, they let players own tradable in-game items with real-world value. Applications include virtual real estate and digital fashion. NFTs challenge traditional copyright laws, making it essential for professionals to understand smart contracts that govern ownership. Overall, NFT rights increase opportunities for creators and firms, fostering innovation and new economic models, while highlighting the importance of tech and legal understanding.

21. Revolutionizing Content Creation and Monetization with Web3. (2024/2025). *BlockApps*. <https://blockapps.net/blog/revolutionizing-content-creation-and-monetization-with-web3/>

Annotation: The content creation world is on the verge of a transformation driven by Web3 technologies, marking a shift towards decentralization, user ownership, and community-driven platforms. Content creators must understand Web3 concepts to navigate and benefit from this evolving landscape.

22. Scarcity-Driven Monetization. (2022/2025 Update). *PMC*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9441551/>

Annotation: Over the past two decades, the U.S. news industry faced a 66% revenue drop, prompting publishers to try new revenue strategies. Some, like paywalls and freemium models, boost income but limit news access, making news scarcer. Others, like reader-focused fundraising, aim to increase revenue organically without reducing news availability. This chapter examines these digital monetization approaches, evaluating the sustainability of scarcity-driven strategies, conditions for profitability, and future industry trends.